

Reverse Hessian Obstructions, Catalecticant Blindness, and a Program Toward VP versus VNP

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Abstract

Let $\det_n$ and per_n denote the determinant and permanent of a generic $n \times n$ matrix, and put

$$D_{\det,n} = \det \text{Hess}(\det)_n, \quad D_{\text{per},n} = \det \text{Hess}(\text{per}_n).$$

We construct a uniform geometric obstruction, based on the Hessian determinant, that separates the Hessian images of the determinant and the permanent under orbit closure: for every $n \geq 3$,

$$D_{\det,n} \notin \overline{\text{GL}_{n^2}(\mathbb{C}) \cdot D_{\text{per},n}}.$$

The proof combines an exact power identity for the determinant, an explicit one-parameter restriction for the permanent, and polystability. In parallel, we certify that ordinary catalecticants do not see this separation for $n = 3$: the complete rank vectors are

$$(1, 9, 45, 165, 270, 270, 165, 45, 9, 1) \quad \text{and} \quad (1, 9, 45, 165, 414, 414, 165, 45, 9, 1).$$

Two independently implemented engines reconstruct the forms and ranks, and an exact gcd certificate proves that $D_{\text{per},3}$ is squarefree.

We then address the relation to the padded permanent used in geometric complexity theory. For a homogeneous degree- d form g in N variables we prove

$$\det \text{Hess}(\ell^k g) = -\frac{k(k+d-1)}{d-1} \ell^{k(N+1)-2} g \det \text{Hess}(g)$$

when the Hessian is taken only in the active variables (ℓ, x) . In the standard GCT ambient space, however, $\ell^{m-n} \text{per}_n$ has unused variables whenever $m > n$, so its full $m^2 \times m^2$ Hessian determinant is identically zero. This yields a rigorous barrier: the ordinary full Hessian determinant alone cannot produce the standard padded-permanent lower bound. No claim that VP differs from VNP is made.

1 Introduction

Valiant's permanent versus determinant problem and the separation of VP from VNP remain central open problems in algebraic complexity [1, 4]. Geometric Complexity Theory (GCT) translates a strengthened form of the problem into non-containment between orbit closures of the determinant and a padded permanent [2, 3]. This translation makes representation theory, invariant theory and algebraic geometry available, but it also exposes exact barriers. In particular, broad occurrence-obstruction programs fail in the relevant asymptotic regime [7, 8], while multiplicity obstructions can still be genuinely stronger in other natural geometric settings [9].

This paper starts from a finite, unpadded Hessian-image separation and asks the question that matters for complexity: can it survive the padding and ambient embeddings that encode determinantal complexity? The answer for the ordinary full Hessian determinant is negative. The positive finite theorem is real and fully explicit, but the same construction collapses to zero on every genuinely padded permanent in the standard ambient space.

The exact contribution is therefore two-sided.

1. We prove a uniform orbit-closure separation after applying the Hessian-determinant covariant in the equal-size unpadded model.
2. We certify a complete ordinary-catalecticant no-go in size three and record finite negative evidence for three shifted-partial-derivative ranks.
3. We prove an active-variable padding factorisation and a full-ambient padding-collapse theorem.
4. We isolate what an anti-padding covariant would have to accomplish before a Hessian program could have consequences for VP versus VNP.

The status labels used throughout are: **PROVED** for a complete paper proof; **CERTIFIED-COMP** for exact reproducible computation; **LITERATURE** for an external theorem used with a precise citation; **CONJECTURAL**, **FAILED**, and **UNKNOWN** otherwise. Finite checks never replace the general proofs.

2 Background on VP, VNP, and GCT

2.1 Permanent, determinant and padding

For a generic $n \times n$ matrix $X = (x_{ij})$, write

$$\det_n(X) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_i x_{i, \sigma(i)}, \quad \text{per}_n(X) = \sum_{\sigma \in S_n} \prod_i x_{i, \sigma(i)}.$$

A determinantal representation of a polynomial f of degree n is an affine linear matrix $A(x)$ of size m such that $f = \det A(x)$. Determinantal complexity is the least such m ; border determinantal complexity allows a limit of such representations. The GCT homogenisation uses the degree- m form

$$\ell^{m-n} \text{per}_n \in \text{Sym}^m(\mathbb{C}^{m^2})^*,$$

where the n^2 permanent variables and a padding variable ℓ are embedded among m^2 ambient variables. The corresponding determinant form is $\det_m \in \text{Sym}^m(\mathbb{C}^{m^2})^*$.

The equal-size comparison between $\det_n$ and per_n in n^2 variables is not the standard padded formulation. Both forms are known to be polystable, and their stabilizers are classical; see [10]. Kumar proves nonnormality phenomena for determinant and padded-permanent orbit closures [11], and the boundary of the 3×3 determinant orbit is explicitly described in [12]. None of these facts may be replaced by an unsupported normality or openness assumption.

2.2 Logical map to VP versus VNP

The relevant implications must be separated by computational model. The permanent family is VNP-complete under p-projections, so

$$\text{per} \notin \text{VP} \iff \text{VP} \neq \text{VNP}.$$

Polynomial determinantal complexity instead characterises the weakly-skew class VP_{ws} (equivalently, polynomial-size algebraic branching programs up to standard polynomial simulations). Consequently, a super-polynomial lower bound for determinantal complexity of the permanent would prove $\text{VP}_{\text{ws}} \neq \text{VNP}$, a major lower bound but not, by itself, the full separation $\text{VP} \neq \text{VNP}$; see [5].

For the border model, the standard GCT implication has the following direction:

$$\ell^{m-n} \text{per}_n \in \overline{\text{GL}_{m^2} \cdot \det_m} \implies C(\ell^{m-n} \text{per}_n) \in \overline{\text{GL} \cdot C(\det_m)} \quad (2.1)$$

for a regular equivariant covariant C with the appropriate character twist. Thus the complexity-relevant target obstruction would be

$$C(\ell^{m-n} \text{per}_n) \notin \overline{\text{GL} \cdot C(\det_m)}. \quad (2.2)$$

The theorem proved in this paper has the opposite orientation, $D_{\det,n} \notin \overline{\text{GL} \cdot D_{\text{per},n}}$, and is therefore not a GCT lower-bound certificate even before padding is introduced. After padding, the full Hessian determinant has the stronger defect $C(\ell^{m-n} \text{per}_n) = 0$, while 0 lies in the closure of every nonzero positive-degree homogeneous orbit. The status map is therefore:

Statement	Status
super-polynomial unrestricted circuit lower bound for $\text{per} \Rightarrow \text{VP} \neq \text{VNP}$	known
super-polynomial determinantal complexity of $\text{per} \Rightarrow \text{VP}_{\text{ws}} \neq \text{VNP}$	known
padded orbit-closure non-containment for all polynomial $m(n) \Rightarrow$ border determinantal lower bound	known in the precise GCT model
reverse unpadded Hessian-image non-containment $\Rightarrow$ padded permanent non-containment	false direction / missing
ordinary full Hessian determinant $\Rightarrow$ padded lower bound	impossible by Theorem 6.3
anti-padding transverse covariant $\Rightarrow$ standard GCT lower bound	unknown

3 The Reverse Hessian Obstruction

Let W be an M -dimensional complex vector space and $f \in \text{Sym}^d W^*$. We use

$$(g \cdot f)(x) = f(g^{-1}x), \quad g \in \text{GL}(W),$$

and write $\mathcal{D}(f) = \det \text{Hess}(f)$ in fixed linear coordinates.

Lemma 3.1 (Hessian covariance). *For every $g \in \text{GL}(W)$,*

$$\mathcal{D}(g \cdot f) = \det(g)^{-2} g \cdot \mathcal{D}(f).$$

In particular, $\mathcal{D}$ is $\text{SL}(W)$ -equivariant.

Proof. The chain rule gives $\text{Hess}(g \cdot f)(x) = g^{-T} \text{Hess}(f)(g^{-1}x)g^{-1}$. Taking determinants proves the claim. $\square$

3.1 Exact determinant identity

Theorem 3.2. *For every $n \geq 2$,*

$$\det \text{Hess}(\det)_n = (n-1)(-1)^{(n-1)(n+2)/2} \det_n^{n(n-2)}.$$

In particular, $\det \text{Hess}(\det_3) = -2 \det_3^3$.

Proof. For $X \in \text{GL}_n$ and $U, V \in M_n$, Jacobi's formula gives

$$d^2(\det)_X(U, V) = \det(X) (\text{tr}(X^{-1}U) \text{tr}(X^{-1}V) - \text{tr}(X^{-1}UX^{-1}V)).$$

At I_n the Hessian bilinear form is $B(U, V) = \text{tr}(U) \text{tr}(V) - \text{tr}(UV)$. At general X , the Hessian is $\det(X)$ times the pullback of B by left multiplication $L_{X^{-1}}$. Since $\det(L_{X^{-1}}) = \det(X)^{-n}$ and the Hessian has size n^2 ,

$$\det \text{Hess}(\det)_n(X) = \det(B) \det(X)^{n^2-2n}.$$

The identity extends polynomially from GL_n to M_n .

On the diagonal subspace, B has matrix $J_n - I_n$ and determinant $(n-1)(-1)^{n-1}$. For every unordered pair $i < j$, the span of E_{ij}, E_{ji} contributes the block $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$, with determinant -1 . Multiplying the $n(n-1)/2$ off-diagonal blocks gives

$$\det(B) = (n-1)(-1)^{n-1+n(n-1)/2} = (n-1)(-1)^{(n-1)(n+2)/2}.$$

□

The power shape is compatible with the general theory of relative invariants and Hessian determinants [14]; the proof above fixes the exact scalar.

3.2 A permanent line restriction

Let $X(t) = J_n + (t-1)E_{11}$, $A = J_n - I_n$, and $C = 0 \oplus (J_{n-1} - I_{n-1})$.

Theorem 3.3 (Exact line restriction). *For every $n \geq 3$,*

$$\det \text{Hess}(\text{per}_n)(X(t)) = \frac{((n-2)!)^{n^2} (n-1)^{2n}}{(n-2)^{(n-2)^2}} (t+n-3)^{(n-2)^2}.$$

Proof. Index the Hessian by pairs (i, j) . Multilinearity gives zero whenever two differentiated variables share a row or column. Otherwise the entry is the permanent of the complementary $(n-2) \times (n-2)$ minor. At $X(t)$ this is $(n-2)!$, except when neither row 1 nor column 1 is removed, where it is $(n-3)!(t+n-3)$. Hence

$$\text{Hess}(\text{per}_n)(X(t)) = (n-2)!A \otimes A + (t-1)(n-3)!C \otimes C.$$

Put $B = A^{-1}C$. Since $A^{-1} = -I_n + \frac{1}{n-1}J_n$, the map B is the identity on

$$U_0 = \{(0, u_2, \dots, u_n) : u_2 + \dots + u_n = 0\},$$

of dimension $n - 2$, and is nilpotent on $\langle e_1, e_2 + \dots + e_n \rangle$. Its eigenvalues are therefore 1 with multiplicity $n - 2$ and 0 with algebraic multiplicity 2. Factoring the Kronecker expression and setting $\alpha = (t - 1)/(n - 2)$ yields

$$\begin{aligned}\det \text{Hess}(\text{per}_n)(X(t)) &= ((n - 2)!)^{n^2} \det(A \otimes A) \det(I + \alpha B \otimes B) \\ &= ((n - 2)!)^{n^2} (n - 1)^{2n} (1 + \alpha)^{(n-2)^2},\end{aligned}$$

which is the stated formula. $\square$

Corollary 3.4. *For $n \geq 3$, $D_{\text{per},n}$ is not a nonzero scalar multiple of an $n(n - 2)$ -th power of a degree- n form.*

Proof. An r -th power has every nonzero line-restriction root with multiplicity divisible by $r = n(n - 2)$. The restriction in Theorem 3.3 has a root of multiplicity $(n - 2)^2$, and $0 < (n - 2)^2 < n(n - 2)$. $\square$

3.3 Polystability and closure

We use the support-cone criterion of [10, Proposition 2.8].

Proposition 3.5. *For every $n \geq 3$, $D_{\text{det},n}$ and $D_{\text{per},n}$ are $\text{SL}(M_n)$ -polystable.*

Proof. Let R be the row-column diagonal torus $E_{ij} \mapsto a_i b_j E_{ij}$ with $\prod_i a_i = \prod_j b_j = 1$. It lies in $\text{SL}(M_n)$ and stabilizes both det_n and per_n , hence both Hessian forms by Theorem 3.1. Its weights on the one-dimensional spaces $\mathbb{C}E_{ij}$ are pairwise distinct, so its centralizer is diagonal.

The group $S_n \times S_n$ acts transitively on the n^2 variables and preserves each Hessian support. Both supports are nonempty by Theorems 3.2 and 3.3. Each Hessian form has degree $q = n^2(n - 2)$. Averaging any support exponent under the transitive permutation group gives $(n - 2)(1, \dots, 1)$; therefore the support cone contains the all-ones vector. The cited criterion gives polystability. $\square$

Lemma 3.6 (Homogeneous GL closure lemma). *Let U be a homogeneous polynomial representation of GL_M whose center generates nonzero scalar multiplication. If nonzero $u, v \in U$ are SL_M -polystable and $u \in \overline{\text{GL}_M \cdot v}$, then $u \in \text{GL}_M \cdot v$.*

Proof. Write $\text{GL}_M \cdot v = \mathbb{C}^*(\text{SL}_M \cdot v)$ and choose $c_j \in \mathbb{C}^*$, $g_j \in \text{SL}_M$ with $c_j g_j v \rightarrow u$. Since u is nonzero and polystable, an SL_M -invariant homogeneous polynomial P of positive degree satisfies $P(u) \neq 0$. Then $c_j^{\deg P} P(v) \rightarrow P(u)$, so $P(v) \neq 0$ and, after taking a subsequence, $c_j \rightarrow c \neq 0$. Thus $g_j v \rightarrow c^{-1}u$. The SL_M orbit of v is closed, so $c^{-1}u \in \text{SL}_M \cdot v$. $\square$

Theorem 3.7 (Uniform reverse-Hessian separation). *For every $n \geq 3$,*

$$D_{\text{det},n} \notin \overline{\text{GL}(M_n) \cdot D_{\text{per},n}}.$$

Proof. By Theorem 3.2, $D_{\text{det},n}$ is an $n(n - 2)$ -th power up to a nonzero scalar. By Theorem 3.4, $D_{\text{per},n}$ is not. Thus they are not in the same $\text{GL}(M_n)$ orbit. If the displayed containment held, Theorems 3.5 and 3.6 would force orbit equivalence, a contradiction. $\square$

Warning 3.8 (Direction mismatch). Theorem 3.7 concerns Hessian images in an equal-size unpadded model and has the reverse orientation from the complexity-relevant obstruction in (2.2). It neither proves a non-containment for the original padded permanent nor gives a circuit, determinantal-complexity, or border- determinantal-complexity lower bound.

4 Reproducible Certification

The new certification chain does not trust historical JSON outputs. It reconstructs all relevant polynomials from monomial data and separates mathematical certificates from machine-dependent runtime logs.

4.1 Two independent catalecticant engines

The primary engine builds $\det_3$, per_3 , their 9×9 Hessian determinants, and ordinary apolar catalecticant matrices. It computes ranks for orders 0 through 4 exactly over $\mathbb{Q}$ using sparse domain matrices; orders 5 through 9 follow from the proved transpose/factorial duality.

The clean-room engine imports no code from the primary engine. It represents polynomials as exponent dictionaries, differentiates them directly, reconstructs both Hessian determinants, builds all ten matrices, and performs its own sparse Gaussian elimination modulo 1009, 32003, and 65521. The orchestration script rejects the run unless polynomial hashes, direct matrix hashes and every modular rank agree with the primary exact rank.

a	$\text{rank } C_a(D_{\det,3})$	$\text{rank } C_a(D_{\text{per},3})$
0	1	1
1	9	9
2	45	45
3	165	165
4	270	414
5	270	414
6	165	165
7	45	45
8	9	9
9	1	1

Table 1: Exact ordinary catalecticant rank vectors.

4.2 Squarefreeness

Lemma 4.1 (Derivative gcd criterion). *Let K be a characteristic-zero field and $0 \neq F \in K[x_1, \dots, x_N]$. If $\gcd(\partial_1 F, \dots, \partial_N F) = 1$, then F is squarefree.*

Proof. If an irreducible polynomial h occurred in F with multiplicity at least 2, writing $F = h^2 q$ would show that h divides every first derivative of F , contradicting the gcd hypothesis. $\square$

A separate deterministic script reconstructs $D_{\text{per},3}$ over $\mathbb{Z}$, computes all nine first partial derivatives, and obtains gcd 1 over $\mathbb{Q}$. By Theorem 4.1, this certifies squarefreeness over $\mathbb{Q}$. Squarefreeness is preserved by the field extension $\mathbb{Q} \subset \mathbb{C}$, hence also holds over $\mathbb{C}$. The integer content gcd of the partial derivatives is 2; the modular data are sanity checks, not the proof.

4.3 Bit-for-bit reproducibility

Certificates contain no absolute paths, elapsed times, hostnames, or timestamps. Running

```
python run_all.py
python scripts/check_reproducibility.py
```

rebuilds the certificates in a fresh temporary directory and compares all eight certificate files byte for byte. `certificates/SHA256SUMS.txt` hashes the mathematical outputs; runtime metadata is isolated under `runtime_logs/`.

5 Catalecticant Blindness

For a degree-nine form $F \in \text{Sym}^9 V^*$, the ordinary apolar catalecticant is

$$C_a(F) : \text{Sym}^a V \longrightarrow \text{Sym}^{9-a} V^*, \quad \theta \longmapsto \theta \cdot F.$$

It is equivariant, and its rank is upper semicontinuous.

Lemma 5.1 (Catalecticant duality). *For every degree- d form F and every $0 \leq a \leq d$,*

$$\text{rank } C_a(F) = \text{rank } C_{d-a}(F).$$

Proof. Use divided-power monomial bases in $\text{Sym}^a V$ and $\text{Sym}^{d-a} V$. The coefficient of a monomial indexed by β in $C_a(F)(\partial^\alpha)$ is the coefficient of the degree- d monomial indexed by $\alpha + \beta$ in F . The same coefficient is the (α, β) entry of the transpose matrix for $C_{d-a}(F)$. In ordinary monomial bases the two matrices differ only by invertible diagonal factorial scalings. Their ranks are therefore equal. $\square$

Theorem 5.2 (Complete ordinary-catalecticant no-go). *Every ordinary catalecticant rank condition that contains $\text{GL}(V) \cdot D_{\text{per},3}$ also contains $D_{\text{det},3}$.*

Proof. For each a , the closed rank locus containing the permanent orbit closure has threshold at least $\text{rank } C_a(D_{\text{per},3})$. The certified table gives

$$\text{rank } C_a(D_{\text{det},3}) \leq \text{rank } C_a(D_{\text{per},3}) \quad (0 \leq a \leq 9).$$

Thus $D_{\text{det},3}$ lies in every such locus and in every intersection of them. $\square$

This theorem concerns rank conditions only. It does not rule out non-rank equations in catalecticant entries, Young or Koszul flattenings, apolar syzygies, singular-locus invariants, or multiplicity obstructions. Such constructions are central in the geometry of secant and complexity varieties [15, 16].

There is also a distinct asymptotic limitation in the literature: Efremenko–Landsberg–Schenck–Weyman prove that shifted partial derivatives alone cannot separate the standard padded permanent from the determinant orbit closure in their stated parameter range [17]. The finite tests below concern the two Hessian-image forms and neither reprove nor strengthen that result.

Three exact shifted-partial tests were also run. Their ranks are

(k, ℓ)	$D_{\text{det},3}$	$D_{\text{per},3}$
$(1, 1)$	65	77
$(2, 1)$	270	369
$(1, 2)$	270	369

and again have the wrong inequality for the desired reverse separation. These are finite CERTIFIED-COMP failures, not a general theorem about shifted partial derivatives.

6 Relation to Padded Permanent Orbit Closures

6.1 What covariance does and does not imply

Because $\mathcal{D}$ is a polynomial covariant up to a determinant character, an original orbit containment forces a target containment after the appropriate twist. A target non-containment can therefore obstruct an original containment when both target values are meaningful. No converse holds: target containment, equality, or vanishing need not reflect the original geometry.

The standard padded permanent exhibits the worst possible loss: its full Hessian determinant is zero.

6.2 Exact active-variable padding formula

Theorem 6.1 (One-padding-variable Hessian factorisation). *Let $g \in \mathbb{C}[x_1, \dots, x_N]$ be homogeneous of degree $d > 1$, let $k \geq 1$, and put $F(\ell, x) = \ell^k g(x)$. Then*

$$\det \text{Hess}_{(\ell, x)}(F) = -\frac{k(k+d-1)}{d-1} \ell^{k(N+1)-2} g(x) \det \text{Hess}_x(g).$$

The identity holds as a polynomial even when $\text{Hess}_x(g)$ is generically singular.

Proof. Put $H = \text{Hess}_x(g)$ and $u = \nabla g$. The full Hessian has block form

$$\begin{pmatrix} k(k-1)\ell^{k-2}g & k\ell^{k-1}u^T \\ k\ell^{k-1}u & \ell^k H \end{pmatrix}.$$

For a scalar a , a column b and a square matrix D , the polynomial block identity

$$\det \begin{pmatrix} a & b^T \\ b & D \end{pmatrix} = a \det D - b^T \text{adj}(D)b$$

holds without any invertibility hypothesis. Factoring the powers of ℓ therefore gives

$$\det \text{Hess}(F) = \ell^{k(N+1)-2} (k(k-1)g \det H - k^2 u^T \text{adj}(H)u).$$

Differentiating Euler's identity $x^T u = dg$ gives $Hx = (d-1)u$. Multiplication by $\text{adj}(H)$ yields

$$\det(H)x = (d-1) \text{adj}(H)u.$$

Taking the scalar product with u and using $u^T x = dg$ gives the polynomial identity

$$u^T \text{adj}(H)u = \frac{d}{d-1} g \det H.$$

Substitution now gives

$$k(k-1) - \frac{k^2 d}{d-1} = -\frac{k(k+d-1)}{d-1},$$

which proves the formula everywhere, including when H is generically singular. $\square$

This formula recovers a canonical trace of $g \det \text{Hess}(g)$ only when the padding line and the active quotient are already marked. It is not, by itself, a regular GL-covariant on unmarked forms.

6.3 Full-ambient collapse

Lemma 6.2 (Unused-variable collapse). *If a polynomial in M ambient variables is independent of one variable, its full $M \times M$ Hessian determinant is zero.*

Proof. The row and column indexed by that variable are zero. $\square$

Theorem 6.3 (Padding-collapse barrier). *Let $m > n$ and regard $F_{n,m} = \ell^{m-n} \text{per}_n$ as a degree- m form in m^2 variables in the standard padded GCT model. Then*

$$\det \text{Hess}_{\mathbb{C}^{m^2}}(F_{n,m}) = 0.$$

Consequently, the ordinary full Hessian-determinant covariant alone cannot separate the padded permanent orbit closure from the determinant orbit closure or imply an asymptotic determinantal-complexity lower bound.

Proof. The form $F_{n,m}$ depends on at most $n^2 + 1$ variables: the n^2 permanent variables and ℓ . Since

$$m^2 - (n^2 + 1) = (m - n)(m + n) - 1 > 0$$

for $m > n$, there is an unused ambient variable. The determinant vanishes by Theorem 6.2.

The image of the padded permanent is therefore the zero form. For any nonzero positive-degree form H , the center of GL_{m^2} scales H , and zero belongs to $\overline{\text{GL}_{m^2}} \cdot H$. Thus the zero target cannot exclude the padded permanent from any determinant target orbit closure. A lower bound would need a different covariant or a canonical active-space refinement. $\square$

7 Beyond Ordinary Catalecticants

The barrier does not rule out differential covariants that retain all Hessian minors, the polar map, Fitting ideals of the derivative module, or a marked transverse Hessian. It only rules out the scalar determinant of the full ambient Hessian.

A viable anti-padding construction must satisfy all of the following:

1. be regular and equivariant on a space stable under the relevant GL_{m^2} action, or explicitly work in a restricted marked model;
2. survive unused ambient variables without making a noncanonical choice;
3. retain a permanent-side invariant that scales uniformly with n, m ;
4. constrain determinant images by a theorem that is stable under orbit closure;
5. evade the known occurrence-obstruction limitations [7, 8]; and
6. be connected by a proved reduction to a standard complexity model.

A first concrete object is the incidence variety of pairs (F, U) with $F \in \text{Sym}^m U^*$ and $U \subset \mathbb{C}^{m^2}$ an active subspace. On a fixed stratum, the maximal nonzero Hessian minors are polynomial, and Theorem 6.1 identifies their marked padding behaviour. The unresolved issue is descent: the minimal active subspace can jump in a degeneration, and choosing it need not be regular.

8 A Candidate Asymptotic Obstruction / Barrier Theorem

The publishable formal result produced by the upgrade is the negative theorem Theorem 6.3. It is more useful than an unsupported positive claim because it removes the ordinary full Hessian determinant from the list of possible direct GCT covariants for the padded model.

Conjecture 8.1 (Canonical active-space extraction). There exists a finite regular equivariant incidence construction whose value on padded-permanent orbits determines the active subspace and the transverse Hessian data of the induced active form.

This conjecture is not equivalent to VP versus VNP. Even if proved, one would still need an asymptotic determinant-versus-permanent separation in its target and a reduction back to the standard unmarked model.

Conjecture 8.2 (Marked transverse-Hessian multiplicity gap). In a suitable parabolic model retaining the padding line, an explicit family of highest weights has a multiplicity gap between transverse-Hessian images of the padded permanent and determinant families.

No partition is presently certified. The conjecture is motivated only by the fact that multiplicity obstructions can be stronger than occurrence obstructions in other natural varieties [9]. Its plausibility is low until a size-3 decomposition is computed exactly.

9 Consequences and Limitations

9.1 Exact consequences

- The reverse Hessian images are separated uniformly in the unpadded same-size model; this is explicitly the opposite orientation from the complexity-relevant padded obstruction.
- All ordinary catalecticant rank tests are blind to that separation for $n = 3$.
- The standard padded permanent is annihilated by the full ordinary Hessian determinant for every genuine padding $m > n$.
- A marked active-variable Hessian retains the factor $g \det \text{Hess}(g)$ with an explicit scalar and padding exponent.

9.2 What is not proved

There is no proof of VP versus VNP, no separation of VP_{ws} from VNP, no unrestricted arithmetic-circuit lower bound, no super-polynomial determinantal- or border-determinantal-complexity lower bound, no padded orbit non-containment obtained from the full Hessian determinant, no Young/Koszul separation, and no coordinate-ring multiplicity gap. The historical degree-six C0 sweep is archived but is not promoted into the claims of this paper.

Recent unconditional lower bounds for symmetric arithmetic circuits show that natural restricted models can be separated: the permanent has exponential lower bounds in the symmetric model of [19], and the homomorphism-polynomial characterisations of [20, 21] develop a broader symmetric algebraic complexity theory. These results do not imply unrestricted VP versus VNP, and the present Hessian barrier should be compared to them only after a precise restricted model is defined. Algebraic-natural-proofs barriers provide a second warning against promoting an efficiently checkable finite property to a general lower-bound method without testing its largeness and usefulness [18].

A Exact Scripts and Certificates

The active files are:

File	Role
scripts/certify_ catalecticant_ranks.py	exact rational primary engine
scripts/independent_ catalecticant_check.py	independent clean-room modular engine
scripts/certify_squarefree_ Dper3.py	exact gcd certificate
scripts/verify_uniform_ formulas.py	finite implementation checks
scripts/transverse_hessian_ experiments.py	exact padding-factor checks
scripts/exploratory_ flattenings.py	finite shifted-partial ranks
scripts/check_ reproducibility.py	isolated byte-for-byte rebuild
certificates/SHA256SUMS.txt	deterministic output manifest

The exact certificate hashes are recorded in `research_notes/RESULT_REPRODUCTION_TABLE.md`. The complete 202-file source archive inventory is in `research_notes/AUDIT_ARCHIVE.md`.

B Literature Audit

The literature review was performed before assigning any novelty status. The distinction between VP, weakly-skew VP, determinantal complexity and the padded border model follows [5]. GCT I/II establish the orbit-closure program [2, 3]; [7, 8] rule out broad occurrence-obstruction strategies, while [9] shows that multiplicity obstructions can be strictly stronger in natural Chow/secant settings; [10] supplies the polystability/support criterion and fundamental invariant context; [11] and [12] constrain claims about determinant and padded-permanent orbit closures; and [13] studies padded polynomial varieties directly. The shifted-partial limitation [17], the algebraic-natural-proofs program [18], and the recent symmetric-circuit results [19–21] delimit the relevant barrier and restricted-model landscape.

No searched source was found to state the exact combination of Theorems 3.3, 3.7, 5.2 and 6.3. Search failure is not proof of novelty. The absolute novelty status is therefore **UNKNOWN** pending specialist review and peer review. Full titles, years, DOIs and URLs are listed in `research_notes/literature_review.md`.

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